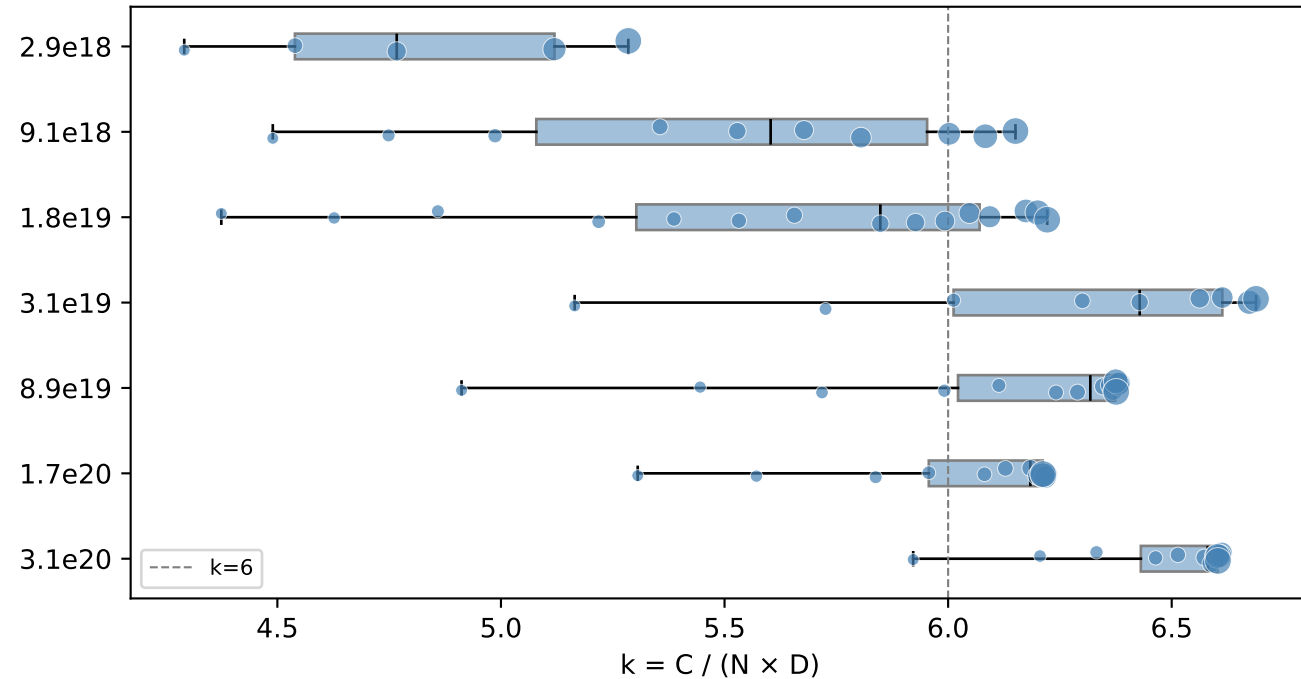


# Empirical FLOP Factors

k by Budget — Delphi



Summary Stats

|             |      |      |      |
|-------------|------|------|------|
| <b>4.80</b> | 4.77 | 4.29 | 5.28 |
| <b>5.48</b> | 5.60 | 4.49 | 6.15 |
| <b>5.61</b> | 5.85 | 4.37 | 6.22 |
| <b>6.24</b> | 6.43 | 5.16 | 6.69 |
| <b>6.09</b> | 6.32 | 4.91 | 6.38 |
| <b>6.03</b> | 6.18 | 5.31 | 6.22 |
| <b>6.47</b> | 6.58 | 5.92 | 6.61 |
| $\mu$       | med  | min  | max  |

**Figure 2:** Empirical FLOP factor  $k = C/(N \cdot D)$  per training budget. Points sized by parameter count  $N$ . The standard  $C = 6ND$  approximation holds at mid-to-high budgets but overestimates at the lowest budget ( $k \approx 4.8$ ).